

INTERNSHIP REPORT

on

DESIGN AND CAD

This internship report is submitted as part of the requirements for
completing the

Bachelor of Technology degree

In

Mechanical Engineering

(Session 2026-2027)

Submitted by

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Under the supervision of

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Certificate.

DECLARATION

I hereby declare that the internship report titled "**SolidWorks CAD Design and Mechanical Modeling Report**" submitted to **Nexonix Reflexo Pvt. Ltd.** and **Rajkiya Engineering College, Banda** is an authentic record of original work carried out by me during the tenure of my internship program.

The matter presented in this report has not been submitted by me for the award of any other degree, diploma, or title at any university or institution.

Signature: _____

Name: Om Nath Kumar Gupta

Date: _____, 2026

ACKNOWLEDGEMENT

I express my sincere gratitude to **Nexonix Reflexo Pvt. Ltd.** for providing me with the opportunity to undertake my internship.

I extend my heartfelt thanks to my company mentor, for their continuous guidance, valuable insights, and constant encouragement throughout the program. I am equally grateful to my faculty mentor, **Mr. Deep Singh Thakur**, for their helpful suggestions and academic supervision.

Lastly, I would like to thank my teammates, institution staff, and family members for their unwavering support and feedback during the project execution.

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Objective:

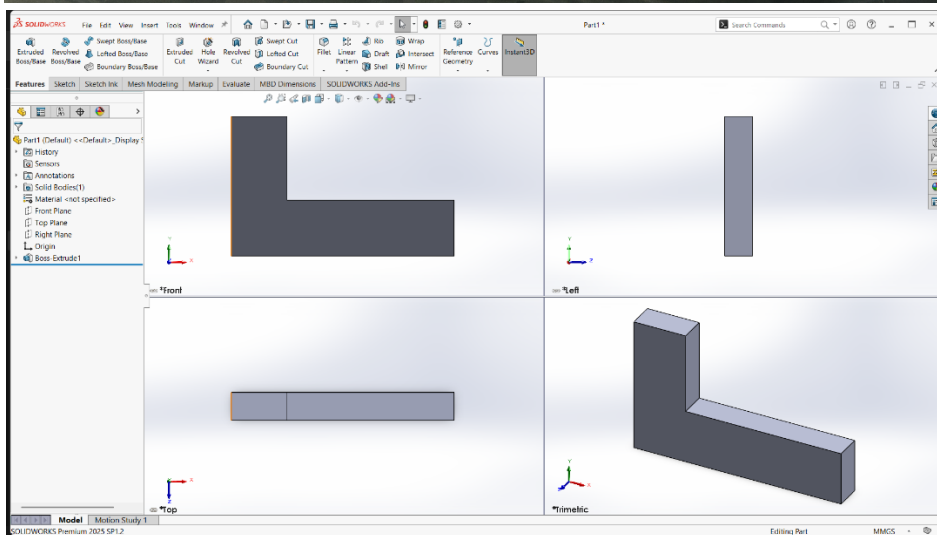
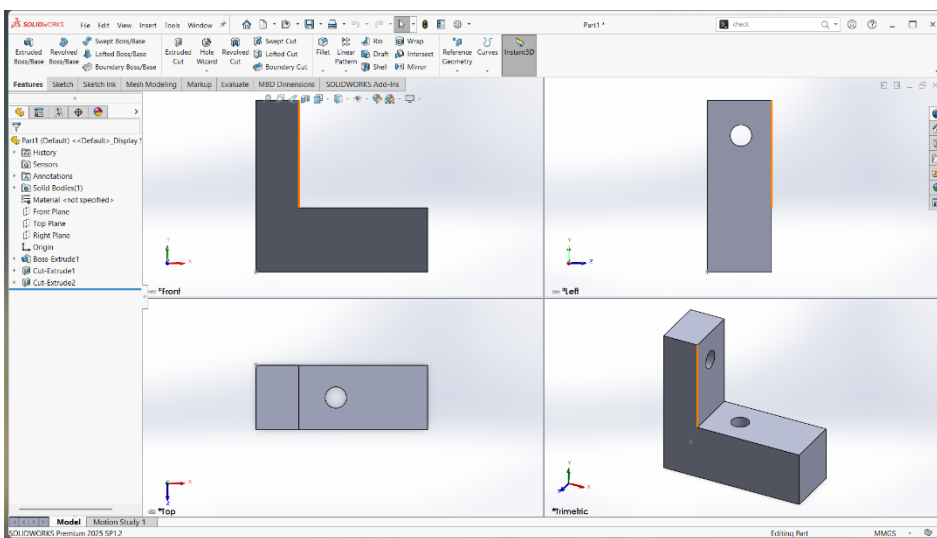
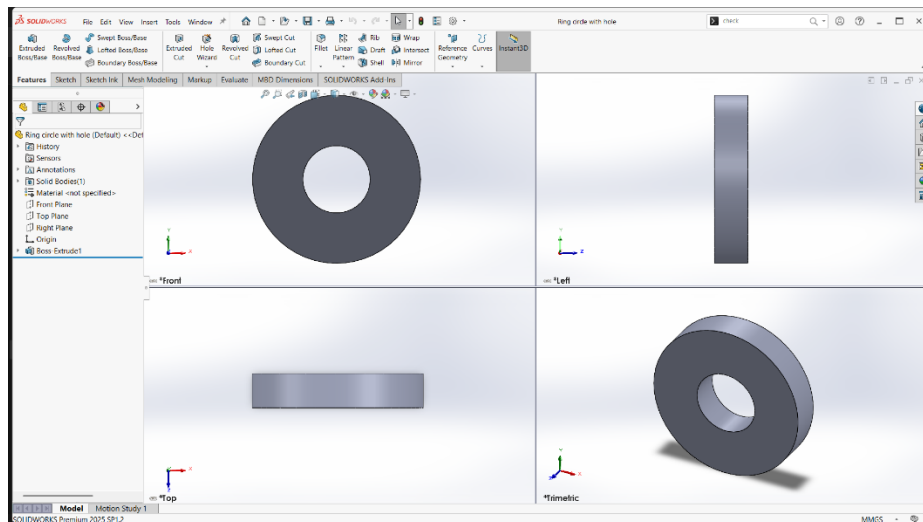
1. To design clean, fully defined 3D models in SolidWorks without any errors in the feature tree.
2. To add real-world manufacturing features like ribs, shell walls, and draft angles so the parts can actually be produced.
3. To fit all the parts together in an assembly and create standard engineering drawings with dimensions and section views.

Methodology:

In this internship, I used SolidWorks to complete the design tasks. I started by creating simple 2D sketches and then converted them into 3D models using basic tools like extrude and revolve. For assembly work, I added the required mates to fit the parts together. This straightforward method helped me understand the basic workflow of SolidWorks and complete each task step by step.

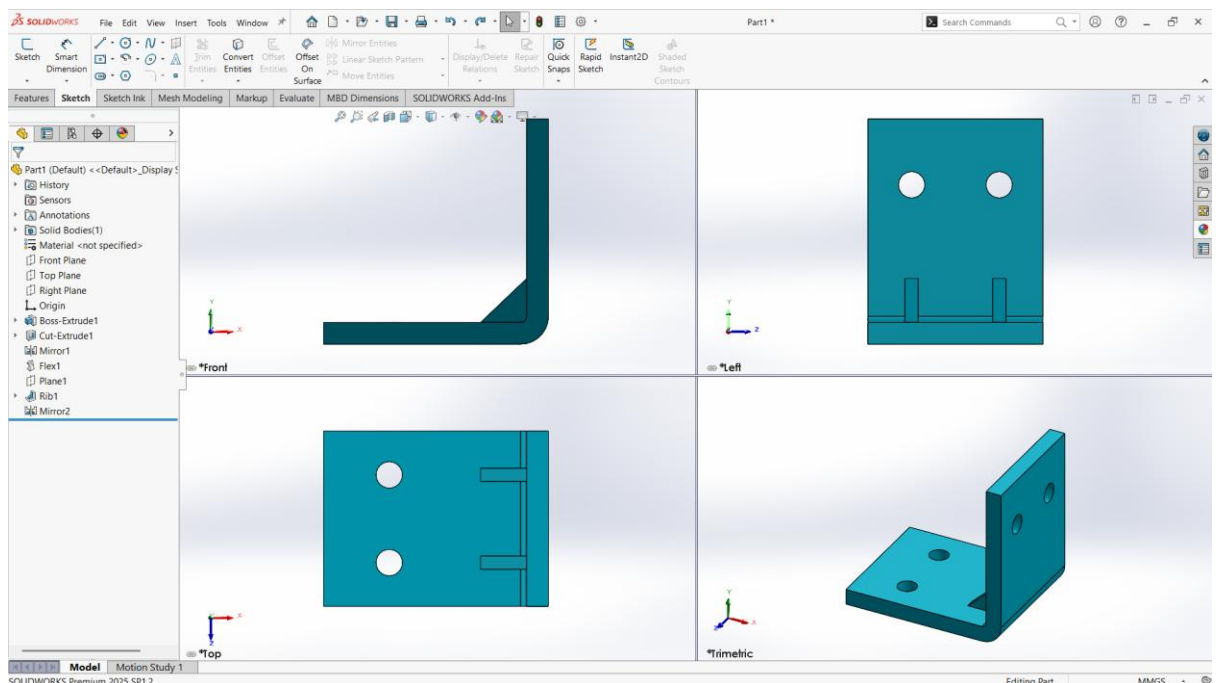
Task 1: Symmetrical Mounting Bracket Methodology

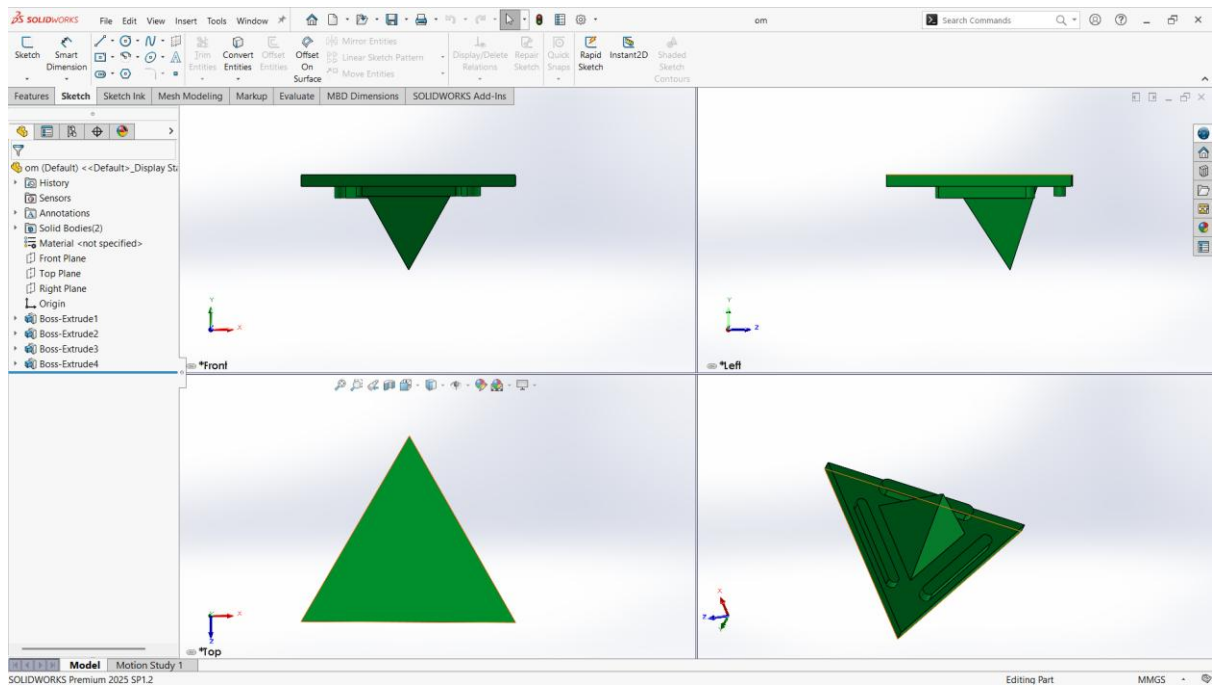
1. Main Body (Boss-Extrude): I created a fully defined sketch of the bracket profile and used Boss-Extrude to form the main body.
2. Mounting Holes (Extruded Cut): I created the required hole sketches and used Extruded Cut to make the mounting holes.
3. Symmetry (Mirror): I used the Mirror feature to duplicate the geometry and maintain perfect symmetry.
4. Support (Rib): I added a Rib feature to strengthen the bent section of the bracket.
5. Bending (Flex): Finally, I used Flex to bend the bracket to the required angle and achieve the final shape.



Task 2: Phone Stand Methodology

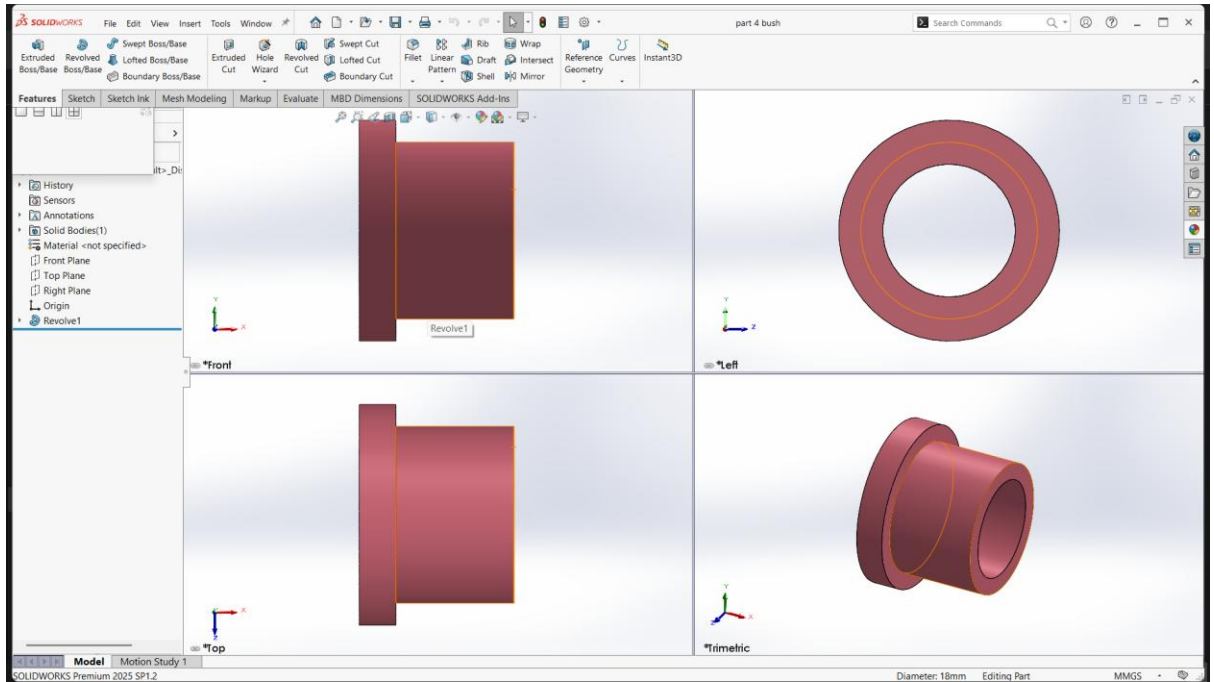
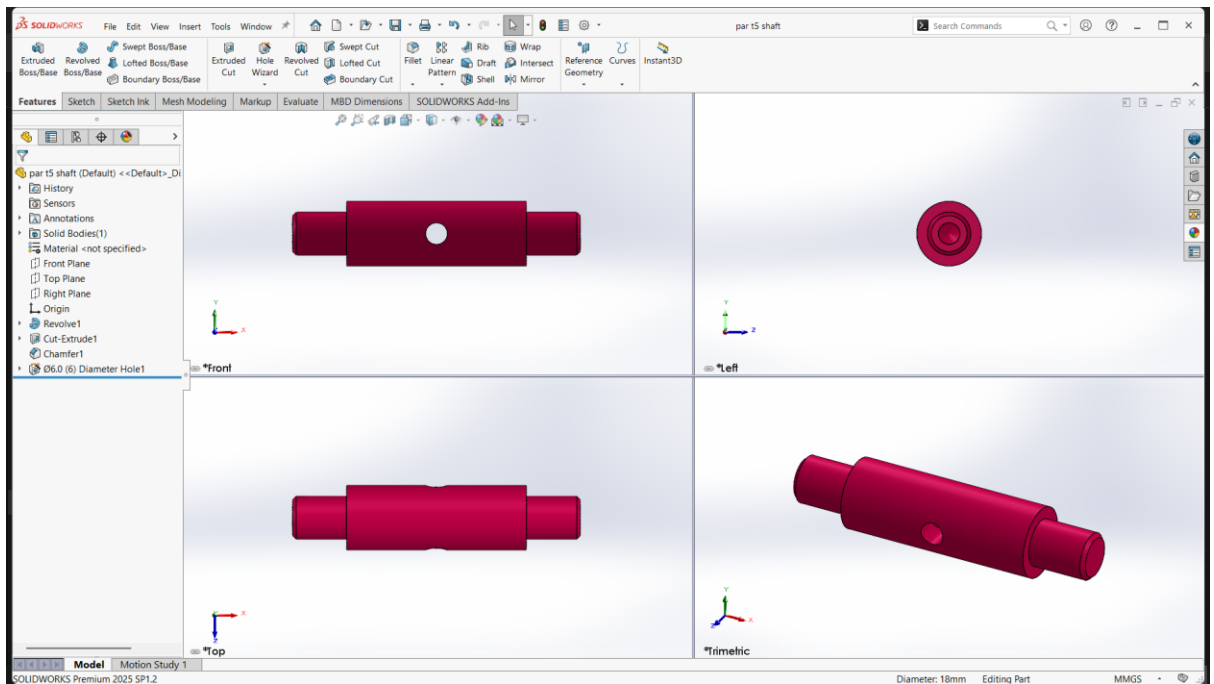
1. **Main Body (Boss-Extrude):** I created a fully defined triangular sketch and extruded it to form the main body.
2. **Repeated Features (Circular Pattern):** I created the required feature and used Circular Pattern to replicate it uniformly.
3. **Hollow Structure (Shell):** I applied Shell to make the stand hollow while maintaining suitable wall thickness.
4. **Manufacturability (Draft):** I added draft angles to make the model easier to manufacture.
5. **Finishing (Fillet):** Finally, I applied Fillet to smooth the sharp edges and improve the overall finish.

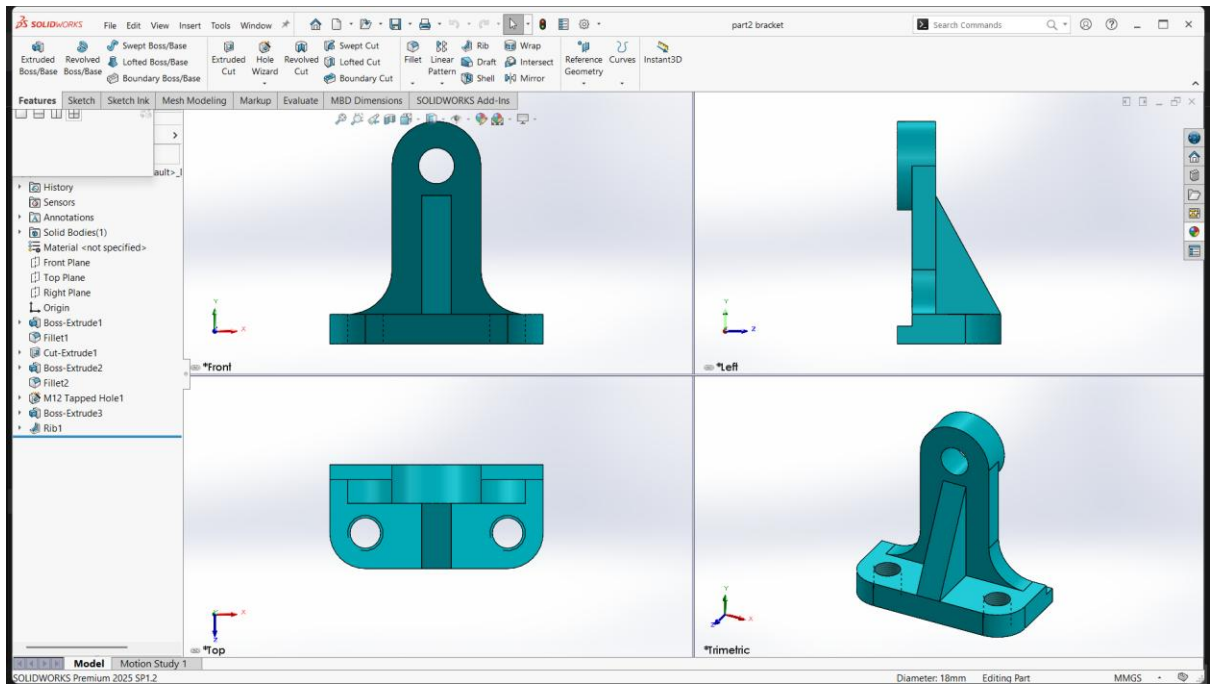
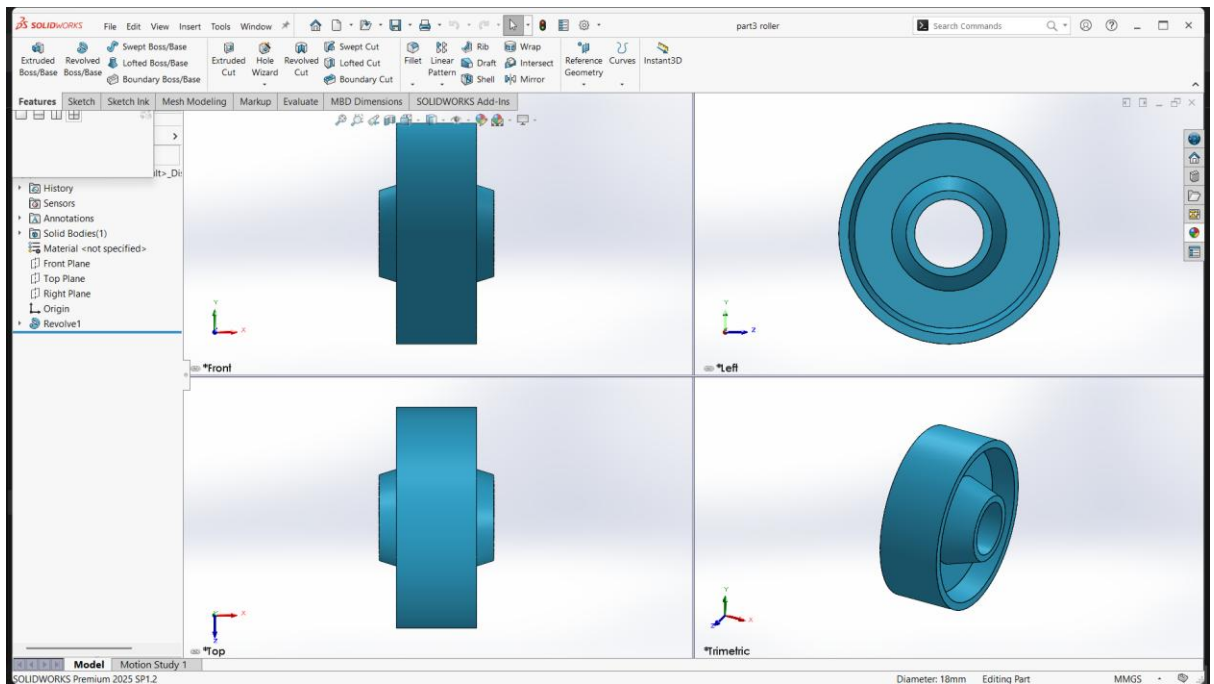


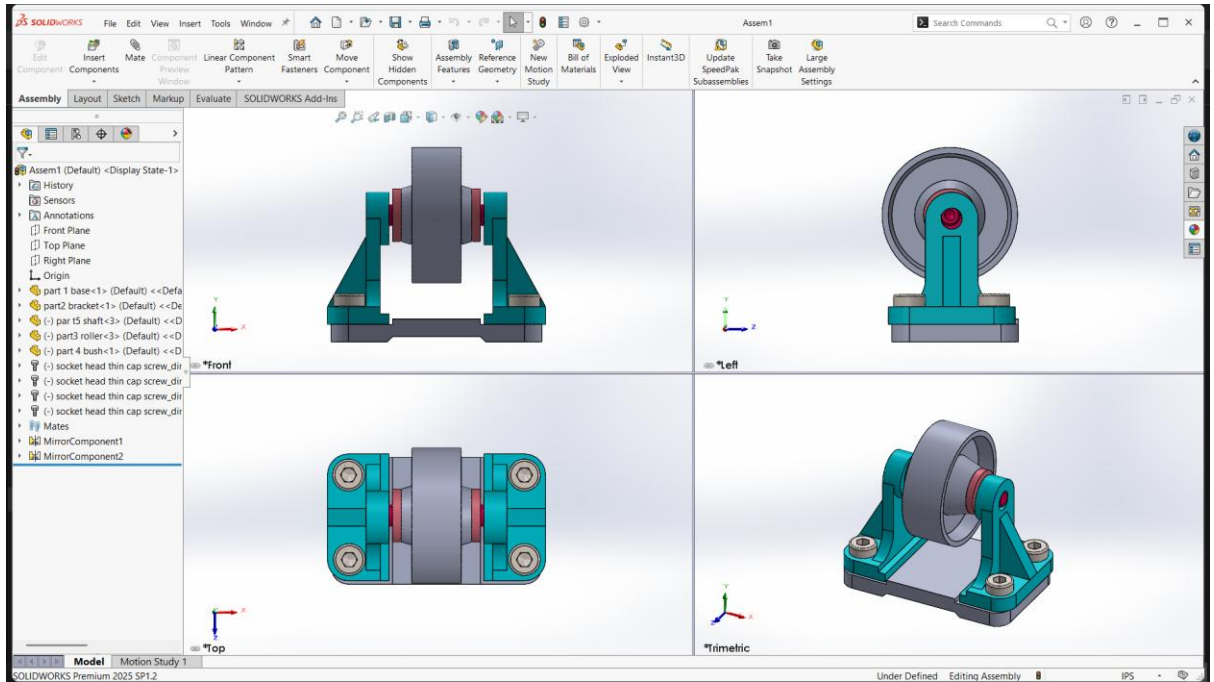
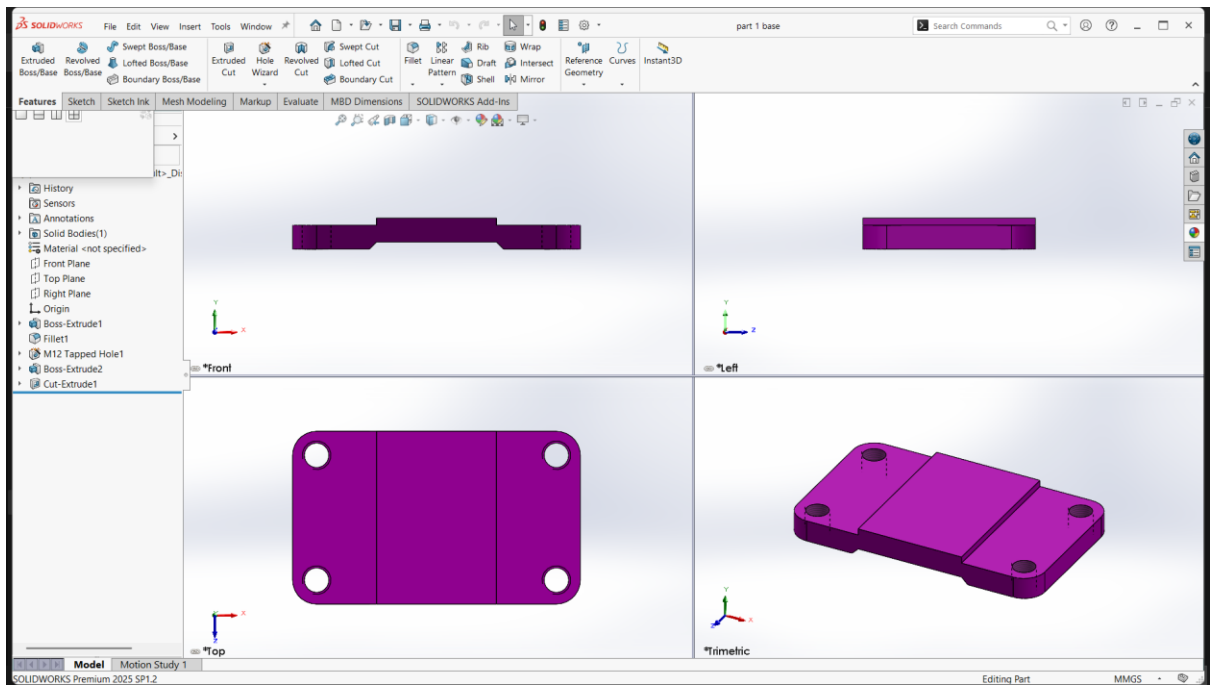


Task 3: Mechanical Assembly

1. **Creating Components:** I created each individual mechanical component separately in SolidWorks with fully defined sketches and appropriate features.
2. **Assembling Components:** I combined the parts into a complete assembly using suitable mates and constraints to ensure proper movement and alignment.
3. **Engineering Drawing:** I generated a production-ready drawing with orthographic views, dimensions, section views, and basic GD&T.
4. **Final Submission:** I checked the assembly and drawing for errors, then exported the required files and uploaded the part, assembly, and drawing files to GitHub along with screenshots on LinkedIn.







RESULT

During my SolidWorks internship, I successfully completed three CAD-based tasks. I designed a symmetrical mounting bracket using features such as Boss-Extrude, Extruded Cut, Mirror, Rib, and Flex. I also created a triangular phone stand using Boss-Extrude, Circular Pattern, Shell, Draft, and Fillet.

For the third task, I created individual mechanical components and assembled them using suitable mates. I also prepared an engineering drawing with orthographic views, dimensions, section views, and basic GD&T.

Through these tasks, I gained practical experience in 3D modelling, assembly, and technical drawing. I also learned to keep sketches fully defined and check models for errors before final submission.

RELEVANCE

The internship was highly relevant to my mechanical engineering studies because it gave me practical experience with CAD modelling and engineering design. I learned how to create parts, modify their geometry, assemble different components, and prepare technical drawings.

The mounting bracket and phone stand helped me understand features such as symmetry, ribs, shells, draft angles, and fillets. The assembly task helped me understand how different parts are connected using mates and how engineering drawings communicate design information.

Overall, the internship improved my SolidWorks skills and gave me a better understanding of how CAD is used in real mechanical design and manufacturing.

LIMITATIONS

The main limitation of the internship was that the work was mostly focused on CAD modelling and did not include physical manufacturing or testing of the designed components. The models were created mainly for learning purposes, so actual factors such as material selection, production cost, strength testing, and durability were not deeply considered.

I also had limited exposure to advanced SolidWorks tools such as Simulation, CAM, and Motion Analysis. However, the internship provided a good foundation in part modelling, assembly, and engineering drawings, which can be developed further through more advanced projects and practical experience.